Steel-Con Steel Construction Systems™

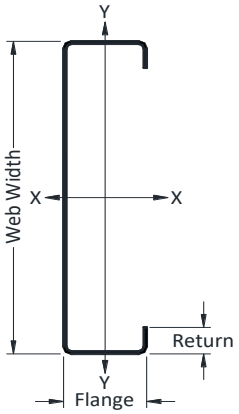
# Product Submittal Sheet

**Orlando Florida Manufacturing Facility**
11250 Astronaut Blvd, Orlando Florida 32837
Phone: 407.438.1664

## 362S162-43

### Product Information

The structural stud is fabricated from prime mill certified steel with a true galvanized coating.
Heavier coatings may be available upon request.




### Steel Material Properties

| | |
|---|---|
| 43 Mil | Labeled Thickness |
| 0.0451" | Design Thickness |
| 0.0428" | Minimum Thickness |
| 33 ksi | Yield Strength (Fy) |
| 45 ksi | Tensile Strength (Fu) |
| G60 | Galvanize Coating Thickness |
| Yellow | Color Code (Painted Ends) |

### Geometric Properties

| | |
|---|---|
| 3-5/8" | Web Width |
| 1-5/8" | Flange Height |
| 1/2" | Return Length |

### LEED - v4 & v4.1 Contributing Credits

All Steel-Con materials have a high inherent recycled steel content and can be used in achieving LEED Certification. Steel-Con's HPD & EPD are available upon request.

- Environmental Product Declaration (EPD) - 1 pt
- Sourcing Raw Material - Up to 2 pts
- Material Waste - 1 pt

### Recycled Content of Steel

- 26.0% Pre-Consumer Scrap Recycled Content
- 8.2% Post-Consumer Scrap Recycled Content
- 34.2% Total Recycled Content

### ASTM and AISI Code Standards

- ASTM A653/A653M, A924/A924M, A1003, C645, C754, C955, C1007
- AISI S100-16 per 2018 IBC, AISI S100-16(2020) w/ S2-20 per 2021 IBC
- 2018, 2021 International Building Codes
- 7th Edition 2020 FBC & FBC-R and 8th Edition 2023 FBC & FBC-R

### Steel-Con Technical Services

For additional information, visit www.steelconsystems.com or contact technical services at 407-404-5292 or Technical@steelconsys.com

### Section Properties

IAPMO UNIFORM ER 0313 | ICC ES ESR-3064P | SSMA

**Table Notes:**

1. The centerline bend radius is based on inside corner radii.
2. Effective properties incorporate the strength increase from the cold work of forming as applicable per AISI S100 A7.2.
3. Tabulated gross properties are based on the full-unreduced cross section of the studs away from punch-out's.
4. For deflection calculations, use the effective moment of inertia.
5. Allowable moment is the lesser of Mal and Mad. Stud distortional buckling is based on an assumed Kϕ = 0.

| Section | Gross Properties | | | | | | | Effective and Distortional Properties | | | | | | Torsional Properties | | | | | | Lu (in) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Area ($in^2$) | Weight (lb/ft) | Ix ($in^4$) | Sx ($in^3$) | Rx (in) | Iy ($in^4$) | Ry (in) | Ixe ($in^4$) | Sxe ($in^3$) | Mal (in-k) | Mad (in-k) | Vag (lb) | VaNet (lb) | Jx1000 ($in^4$) | Cw ($in^6$) | Xo (in) | m (in) | Ro (in) | ß | |
| 362S162-43 | 0.340 | 1.16 | 0.710 | 0.392 | 1.445 | 0.127 | 0.611 | 0.710 | 0.372 | 7.34 | 7.62 | 1739 | 676 | 0.23 | 0.376 | -1.297 | 0.782 | 2.036 | 0.594 | 42.5 |

### Limiting Wall Heights

**Table Notes:**

1. Listed wind pressures represent calculated designed wind pressure (0.6 W based on ASCE 7). For deflection calculations, listed wind pressures have been reduced by 0.70 as allowed by IBC and FBC. The 5 psf pressure has not been reduced for deflection checks.
2. Studs must be braced against rotation and lateral movement at all supports.
3. Studs are assumed to be adequately braced at a maximum spacing of Lu to develop full allowable moment.
4. Web crippling check is based on 1" of bearing at end supports and 3" of bearing at interior support.
5. Shear and web crippling capacity at end supports have not been reduced for punch-out's. Shear and web crippling capacity at interior support have been reduced for the presence of punch-out adjacent to the support.
4. Combined bending and shear check at interior support is based on unreinforced web per AISI S100 (Eq. H2-1). Shear capacity and combined bending and shear check at interior support have been reduced for the presence of punch-out's adjacent to support.

| Stud Spacing (in) | Non-Composite Fully Braced (5 psf) | | | Non-Composite Fully Braced (15 psf) | | | Non-Composite Fully Braced (20 psf) | | |
|---|---|---|---|---|---|---|---|---|---|
| | L/120 | L/240 | L/360 | L/240 | L/360 | L/600 | L/240 | L/360 | L/600 |
| 12" o.c. | 26' 6" | 21' 0" | 18' 4" | 16' 5" | 14' 4" | 12' 1" | 14' 11" | 13' 0" | 11' 0" |
| 16" o.c. | 24' 1" | 19' 1" | 16' 8" | 14' 11" | 13' 0" | 11' 0" | 13' 7" | 11' 10" | 10' 0" |
| 24" o.c. | 21' 0" | 16' 8" | 14' 7" | 12' 9" | 11' 5" | 9' 7" | 11' 1" | 10' 4" | 8' 9" |